

Chern++: computational results on positivity for Morin Thom polynomials

Abstract

We report on a machine-checked study of the two positivity conjectures for Morin singularities. The infrastructure computes $\mathcal{Q}_d$ for $d \leq 7$ from the explicit orbit equations of Bérczi–Szenes §7.3, self-checked against every invariant fixed in advance and agreeing with Rimányi’s published tables wherever those reach. On top of it: Rimányi’s conjecture is verified for A_6 (first time), the two reductions of the A_5 handoff note are shown to survive at $d = 6$ but to *fail* at $d = 7$ – while Rimányi’s conjecture itself survives there, so the reduction machinery, not the conjecture, is what is specific to small d — the proposed prefix-positivity route is shown to fail already at $d = 6$, and a notion of *denominator certificate* is introduced whose successes are proofs and whose failures are theorems. The certificate machinery reproduces the $d = 4$ strong positivity theorem and Proposition 3 of the handoff note mechanically. All coefficients agree with Rimányi’s independently computed tables wherever those reach, including A_7 . Notation follows `rimanyi_positivity.pdf`, `report.pdf` and `a5_weak_positivity_handoff.pdf`, cited as [N], [R], [H].

1 Setting and notation

For the Morin singularity A_d , the Bérczi–Szenes formula expresses Tp_{A_d} as an iterated residue whose only d -dependent input is $\mathcal{Q}_d = \mathrm{mdeg}_{T_d}(\mathcal{O}_d)$, the equivariant multidegree of the Borel orbit closure $\mathcal{O}_d = \overline{B_d \cdot \epsilon_{\mathrm{ref}}}$ inside $\widehat{N}_d \subset \mathrm{Hom}(\mathbb{C}^d, \mathrm{Sym}^2 \mathbb{C}^d)$. Normalising $z_d = 1$ and setting $x_j = z_j/z_{j+1}$, positivity is read off the chamber series

$$F_d(x) = \frac{\prod_{m < l} (1 - z_m/z_l) \mathcal{Q}_d}{\prod_{m+r \leq l} (1 - z_m/z_l - z_r/z_l)} = \frac{N_d(x)}{\prod_r (1 - f_r(x))} = \sum_{\beta \geq 0} A_\beta x^\beta,$$

where every f_r has nonnegative coefficients and zero constant term. Write $\tau(i, j, k, \dots) = (j - i, j, k, \dots)$ for the first adjacent transposition, and $C(M)$ for the ℓ -free Chern coefficient of [R, Thm. 5].

2 Infrastructure

2.1 Computing $\mathcal{Q}_d$

We take the route of [N, §7.3]: rather than parametrise the orbit and eliminate the group variables, we write down the quadratic relations of Proposition 7.3 directly, saturate by the defect-zero coordinates to remove the spurious components that appear from $d = 6$ on ([N, §7.4]), and read the multidegree off an initial ideal.

The design point worth stating is that every quantity the computation can be checked against is known in advance, and all of them are checked: multihomogeneity of each relation under the torus weights; the codimension $\dim \widehat{N}_d - \binom{d}{2}$ forced by homogeneity of the residue formula; vanishing on random B_d -translates of ϵ_{ref} ; $\deg \mathcal{Q}_d$; exactness of the chamber division; and the normalisation fixing the c_1^d coefficient. The stage refuses to emit an artifact otherwise. Externally, the resulting Thom polynomials agree with Rimányi’s published tables at every order and relative dimension those cover (§4).

2.2 Reaching $d = 7$

Three observations bring $d = 7$ into range; the arithmetic they save is recorded in the source rather than here.

Saturation is a division. For a homogeneous ideal and degrevlex with v last, a Gröbner basis saturates by v on dividing each element by the largest power of v dividing it. This replaces the elimination that a general-purpose saturation performs — and which does not terminate here — with one basis computation per defect-zero coordinate. Both the placement of those coordinates and the sequence in which they are moved to the end matter by two orders of magnitude, for the same ideal and the same answer.

Every one of those saturations is needed. If $\text{in}(I) : v = \text{in}(I)$ then $I : v = I$, since for $f \in I : v$ we have $v \cdot \text{in}(f) = \text{in}(vf) \in \text{in}(I)$, and $I \subseteq I : v$ then forces the Hilbert functions equal. The left-hand condition is combinatorial, so it can be tested on the initial ideal alone after each step. It never holds until the last coordinate is saturated.

The two phases after the basis are combinatorics, not algebra. The initial ideal is monomial, and $P_S = (x_i : i \in S)$ contains a monomial ideal exactly when S meets the support of every generator; the minimal primes are therefore the minimal *transversals* of those supports. Localising at one is a restriction of exponent vectors, and the multiplicity is the number of monomials outside the resulting artinian ideal. Neither step needs a primary decomposition, and bounding the transversal search at the expected codimension is sound, since every subset of a transversal of that size is itself of that size — which leaves the codimension check intact in both directions.

One consequence is worth stating, because it is easy to mistake for a discrepancy. The variable order changes the degeneration but not what it computes: at $d = 7$ two orders give 621 and 572 components, with different multiplicities, summing to the same $\mathcal{Q}_7$. The component structure is an artefact of the term order; the multidegree is an invariant of the variety. A related null result: over $\mathbb{Q}$, $\text{GF}(32003)$ and $\text{GF}(2^{31} - 1)$ the initial ideals are identical and the cost is the same to within a few percent, so coefficient growth is not what this computation is spending on.

2.3 Scale of the objects

Table 1 is the practical reason sampling is no substitute for a proof: roughly half the coefficients of N_d are negative at every order, and the counts grow fast. Two features of it are structural rather than incidental. The Vandermonde $\prod_{m < l} (1 - z_m/z_l)$ collapses to exactly $d!$ terms in chamber coordinates — it is a Weyl denominator, so the $2^{\binom{d}{2}}$ products cancel down to a sum over S_d — and $\deg N_d$ is $\binom{d}{2}$ more than $\deg \mathcal{Q}_d$ measured in the same coordinates.

Table 1: The mined chamber algebras. $\deg \mathcal{Q}_d = \dim \hat{N}_d - \binom{d}{2}$ is the codimension of the orbit closure; there are $d - 1$ chamber variables throughout.

	$\dim \hat{N}_d$	$\deg \mathcal{Q}_d$	$\mathcal{Q}_d$ terms	$\max \mathcal{Q}_d $	Vand. terms	N_d terms	$\max N_d $	N_d negative
A_1	0	0	1	1	1	1	1	0 (0%)
A_2	1	0	1	1	2	2	1	1 (50%)
A_3	3	0	1	1	6	6	1	3 (50%)
A_4	7	1	3	2	24	54	3	27 (50%)
A_5	13	3	19	8	120	814	18	414 (51%)
A_6	22	7	418	233	720	26618	800	13309 (50%)
A_7	34	13	14586	37880	5040	1210404	196803	605234 (50%)

Table 2 records the orbit closure itself. The degree there is three agreeing computations, not one: $\mathcal{Q}_d(1, \dots, 1)$, the Hilbert series of the flat degeneration, and the sum of the component

multiplicities. Only the component count is not an invariant — it belongs to the Gröbner degeneration, and a different term order gives a different one summing to the same class.

Table 2: The orbit closure $\mathcal{O}_d \subset \widehat{N}_d$. “mult” is the range of component multiplicities, “deg Hilb” the degree of the Hilbert numerator. For $d \leq 3$ the orbit fills $\widehat{N}_d$ and there is no degeneration.

	$\dim \widehat{N}_d$	$\dim \mathcal{O}_d$	codim	deg $\mathcal{O}_d$	components	mult	deg Hilb	$\mathcal{Q}_d$
A_1	0	0	0	1	—	—	0	1
A_2	1	1	0	1	—	—	0	1
A_3	3	3	0	1	—	—	0	1
A_4	7	6	1	2	2	1–1	2	irreducible
A_5	13	10	3	6	5	1–2	5	$1 \cdot 2$
A_6	22	15	7	55	44	1–2	12	irreducible
A_7	34	21	13	957	572	1–8	21	irreducible

2.4 How much cancellation is being asked for

Strong Laurent positivity fails from $d = 5$, so some A_β are negative; the weak conjecture survives because each Chern coefficient sums many of them. Table 3 measures the gap.

Table 3: Negative Laurent coefficients against the positive mass they sit in, on the truncation shown, and the Chern coefficients at $\ell = 0$ they cancel into.

	$\deg \leq$	A_β	negative	%	least	neg. mass	of pos.	$\min C(M)$
A_4	12	305	0	0.0%	—	0	0.00%	1
A_5	12	687	7	1.0%	−1	7	0.03%	1
A_6	10	526	6	1.1%	−1	6	0.09%	1
A_7	10	657	13	2.0%	−2	14	0.19%	1

Three things stand out. The negative coefficients are *rare*, growing from 1% to 2% of the terms. They are *flat*: every one is −1, with a single −2 at $d = 7$, while the positive coefficients reach 10^5 and beyond. And the mass they carry is a fraction of a percent of the positive mass. Meanwhile $\min C(M) = 1$ at every order and every relative dimension we reach — the Chern coefficients are not merely nonnegative but strictly positive, and never larger than they need to be at the bottom.

This is the shape of the open question in §7. A bound on the negative part of $C(M)$ in terms of its largest term would settle the conjecture without any pairing, and the numbers say such a bound has a great deal of room in it.

2.5 Exactness

No result below depends on floating point. One point deserves emphasis because it is a trap: Laurent coefficients are accumulated in `int64`, and they outgrow it. A wrapped accumulator reads as a large *negative* number, that is, as a spurious counterexample to the very conjecture under test. The evaluator therefore replays the fixed point in floating point as a magnitude probe and raises rather than returning wrapped values. Beyond that point the expansion is run in residue arithmetic instead; see §3.

3 Results for A_6

Proposition 1 (verified). *Rimányi’s conjecture holds for A_6 at every relative dimension $\ell \leq 7$, and for A_5 at $\ell \leq 11$.*

These are finite exact computations, not proofs of the infinite statement. For A_5 this matches the range of [R, Thm. 8]; the A_6 verification is new. Past $\ell = 5$ the coefficients outgrow `int64` — they reach 1.75×10^{22} at A_6 , $\ell = 7$ — so the expansion is run modulo several word-sized primes and reconstructed by CRT, with one prime held back and spent on verifying the reconstruction. The grouping into Chern monomials is computed from the grid geometry rather than from which cells are nonzero, so that the residues of a given coefficient correspond across primes; grouping by value would misalign them, since a nonzero coefficient can vanish modulo one prime and not another.

3.1 External agreement

Rimányi’s tables at `tpp.web.unc.edu` give the Thom polynomials at relative dimension one, computed by the restriction-equation method. All 15, 30 and 58 coefficients for A_4 , A_5 , A_6 agree with ours. Since that method is independent of the residue formula implemented here, this constrains $\mathcal{Q}_d$ externally rather than self-consistently, and is a considerably sharper check than the eleven classical values at $\ell = 0$.

The negative Laurent coefficients of F_6 behave much as at $d = 5$; see Table 4. The first counterexample to strong Laurent positivity occurs at $d = 6$ at the same β as at $d = 5$, extended by a zero. The last column is the one that matters in §3.3.

Table 4: Negative Laurent coefficients of the chamber series, to total degree 11.

	terms	negative	first negative	most negative	with $i = 0$
A_4	237	0	—	—	0
A_5	490	6	(1, 1, 2, 1)	−1	0
A_6	796	10	(1, 1, 2, 1, 0)	−2	2

3.2 The reductions of [H] at $d = 6$

Observation 2. *Both reductions of [H] survive at $d = 6$ throughout the tested range ($\deg \leq 11$): the unpaired tail $i > j \Rightarrow A_{i,j,\dots} \geq 0$, and the paired inequality $A_{i,j,\dots} + A_{j-i,j,\dots} \geq 0$ for $0 \leq i \leq j$.*

This is evidence that the τ -pairing frame is structural rather than an A_5 accident.

3.3 Prefix positivity fails at $d = 6$

[H, §8] observes that every tested coefficient of $B = F_5/(1 - a)$ is nonnegative, and [H, §11.4] proposes proving that as a stepping stone, since $B_{i,j,k,l} = \sum_{r \leq i} A_{r,j,k,l}$ turns the paired inequality into a statement about adjacent differences of a prefix array.

Proposition 3. *$F_6/(1 - a)$ is not coefficientwise nonnegative. Indeed $A_{(0,2,3,2,2)} = -1$.*

Proof. The prefix sum runs over $r \leq i$, so $B_{0,j,k,l} = A_{0,j,k,l}$ and any negative coefficient with $i = 0$ survives untouched. F_5 has no negative coefficient with $i = 0$ in the tested range; F_6 has two, the first being $A_{(0,2,3,2,2)} = -1$. Confirmed along two independent code paths, truncated series products in exact integers and the XLA fixed-point grid. $\square$

The route of [H, §11.4] is therefore specific to $d = 5$. Note that the obstruction is at the very first level of the prefix sum, so no refinement of the argument recovers it.

4 A_7 : the reductions break, the conjecture does not

$\mathcal{Q}_7$ is computable — see §2.2 — and it changes the picture. It is worth being precise about what fails, because the two are easy to conflate. Strong Laurent positivity has been false since $d = 5$ and is no more false at $d = 7$. Rimányi’s conjecture, the actual target, *survives*: every Chern coefficient of Tp_{A_7} is positive at every relative dimension we reach. What breaks is the *reduction machinery* of [H] — the two inequalities through which the A_5 programme runs.

4.1 Validation first

Because the claim is a negative one, the input has to be beyond question. Our Tp_{A_7} reproduces Rimányi’s published tables exactly: all 15 coefficients at $\ell = 0$ and all 105 at $\ell = 1$, and likewise for A_4 , A_5 , A_6 . Those tables are maintained by Rimányi himself at tpp.web.unc.edu and are computed by the restriction-equation method — an approach independent of the residue formula implemented here — so the agreement constrains $\mathcal{Q}_7$ externally rather than self-consistently.

We note the limit of that check. The companion page tpp.web.unc.edu/tp-linfinity/ tabulates the rational functions F_Q , whose numerators are the closest published analogue of our $\mathcal{Q}_d$; for A_4 it gives numerator $2z_1 + z_2 - z_4$, exactly our $\mathcal{Q}_4$. But that page stops at $\mu = 6$: **no A_7 data appears there**, so $\mathcal{Q}_7$ itself has no published counterpart to compare against. The Thom polynomial tables are therefore the strongest external constraint available, and they are what we use.

Internally, every offending coefficient below agrees between two unrelated code paths — the truncated series in exact Python integers and the XLA fixed-point grid — and all of them sit at total degree 7 to 10, inside the truncation degree, so they are exact rather than artefacts of a cut-off.

4.2 How the reductions fail

Observation 4. *The unpaired tail fails at $d = 7$: $A_{(2,1,1,2,2,1)} = -1$ with $i = 2 > j = 1$. Proposition 3 of [H] proves this cannot happen at $d = 5$, and it does not happen at $d = 6$.*

Observation 5. *The paired inequality fails at $d = 7$, and does so in the sharpest possible way. Take $\beta = (1, 2, 1, 2, 2, 1)$. Then $j = 2i$, so $\tau(\beta) = \beta$: this is a fixed point of the transposition, where the paired inequality reads $2A_\beta \geq 0$. But $A_\beta = -1$. There is no partner to appeal to.*

4.3 Why Rimányi’s conjecture survives it

The mechanism is the one that saved $d = 5$, and it is worth seeing at $d = 7$ concretely. In the α coordinates $\beta = (1, 2, 1, 2, 2, 1)$ is $\alpha = (1, 1, -1, 1, 0, -1, -1)$, so it contributes to the Chern monomial indexed by the multiset $M = \{1, 1, 1, 0, -1, -1, -1\}$ — at $\ell = 0$, the monomial $c_1 c_2^3$. That multiset has 35 ballot orderings, of which eight carry a nonzero coefficient:

Five negative units against forty positive: $C(M) = 35 > 0$. And 35 is precisely the coefficient of $c_1 c_2^3$ in Rimányi’s published Tp_{A_7} at $\ell = 0$, so the cancellation is confirmed against his table and not only against our own arithmetic. The unpaired-tail violation behaves the same way: $A_{(2,1,1,2,2,1)} = -1$ sits inside $M = \{2, 1, 0, 0, -1, -1, -1\}$, the monomial $c_1^2 c_2 c_3$, whose coefficient is 301.

So $d = 7$ separates two things that $d \leq 6$ kept together. The conjecture is about $C(M)$, and $C(M)$ is fine. The reductions are about the individual A_β and their τ -orbits, and those are not. The τ -pairing is a way of certifying the sum by controlling its summands two at a time; at $d = 7$ the summands are no longer controllable that way, while the sum is unharmed.

Table 5: The Chern coefficient $C(M)$ for $M = \{1, 1, 1, 0, -1, -1, -1\}$ at $d = 7$. The offending β of the second observation is the third row.

α	β	A_β
$(1, -1, 1, 1, 0, -1, -1)$	$(1, 0, 1, 2, 2, 1)$	1
$(1, 0, 1, 1, -1, -1, -1)$	$(1, 1, 2, 3, 2, 1)$	4
$(1, 1, -1, 1, 0, -1, -1)$	$(1, 2, 1, 2, 2, 1)$	-1
$(1, 1, 0, 1, -1, -1, -1)$	$(1, 2, 2, 3, 2, 1)$	-4
$(1, 1, 1, -1, -1, -1, 0)$	$(1, 2, 3, 2, 1, 0)$	5
$(1, 1, 1, -1, -1, 0, -1)$	$(1, 2, 3, 2, 1, 1)$	6
$(1, 1, 1, -1, 0, -1, -1)$	$(1, 2, 3, 2, 2, 1)$	6
$(1, 1, 1, 0, -1, -1, -1)$	$(1, 2, 3, 3, 2, 1)$	18
$C(M)$		35

4.4 Consequences

The τ -pairing frame is therefore $d \leq 6$, not structural. Anything built on it — the unpaired-tail reduction, the paired series $\mathcal{P}$ of [H, eq. (12)], the $J_d(1/2, \cdot)$ analysis of §6 — is an argument about small d and cannot be the shape of a general proof. That is a strong constraint on where to look next, and it is the main reason we report the failure rather than the verification.

5 Denominator certificates

Definition 6. Let $N/\prod_{r=1}^R(1 - f_r)$ with each f_r nonnegative and of zero constant term. An *order- k certificate* is an identity

$$N = \sum_{|S| \leq k} P_S \prod_{r \in S} (1 - f_r), \quad \text{every } P_S \text{ coefficientwise nonnegative,}$$

over subsets $S \subseteq \{1, \dots, R\}$.

Dividing through, $F = \sum_S P_S / \prod_{r \notin S} (1 - f_r)$ is a sum of products of nonnegative series, so $F \geq 0$. Order 0 is the statement that N itself is nonnegative. [H, §10.3] reports a first-order search of this shape coming back infeasible.

5.1 Why a returned certificate is a proof

$P_\emptyset$ is not a free unknown: since $\prod_{r \in \emptyset} (1 - f_r) = 1$, it is the remainder $N - \sum_{S \neq \emptyset} P_S \prod_{r \in S} (1 - f_r)$. We therefore solve a feasibility LP in the nonempty parts only, snap the solver’s floating-point answer to exact rationals, and recompute $P_\emptyset$ in exact arithmetic. The identity then holds by construction, and the single remaining question, whether that exact remainder is coefficientwise nonnegative, is decided exactly. The LP is demoted to a search heuristic; correctness never depends on it. The output is a finite algebraic identity checkable in any computer algebra system.

5.2 Why a failure is a theorem

Lemma 7 (projection). *A constraint on a monomial of total degree T involves only the coefficients of the P_S of degree $\leq T$, because every $\prod_{r \in S} (1 - f_r)$ has constant term 1 and no negative-degree terms. Truncating both the constraints and the unknowns at degree T is therefore an exact projection of the feasible set, not a further restriction.*

Consequently infeasibility of the truncated program proves that no certificate of that order exists at *any* degree. Failed searches become impossibility results.

5.3 A_4

Proposition 8. F_4 admits an order-4 certificate with $\deg P_S \leq 8$, exhibited explicitly and verified exactly; and no certificate of order ≤ 3 exists at any degree. The minimal order is 4.

This is an independent computational proof of [N, Thm. 1] by a method using no special feature of $d = 4$. The lower bound already follows by hand at depth 1: writing m_S for the constant term of P_S , matching the constant term of $N = 1 - a - b - c + \dots$ forces $\sum_S m_S \leq 1$, while matching the coefficients of a, b, c forces

$$\sum_{S \ni f_0} m_S \geq \frac{1}{2}, \quad \sum_{S \ni f_2} m_S \geq 1, \quad \sum_{S \ni f_5} m_S \geq 1,$$

where $f_0 = 2a$, $f_2 = b + ab$, $f_5 = c + abc$ are the only factors containing the respective bare monomials. With a budget of 1 these can only be met if a single S contains all three. The machine-found certificate saturates all four bounds exactly, its only two constant terms being $\frac{1}{2}(1 - f_0)(1 - f_2)(1 - f_4)(1 - f_5)$ and $\frac{1}{2}(1 - f_2)(1 - f_4)(1 - f_5)(1 - f_6)$.

5.4 A_5 : how far the obstruction reaches

Proposition 9. No denominator certificate of order ≤ 7 exists at any degree for F_5 , nor for $F_5/(1 - a)$.

For F_5 this is expected, since $F_5 \not\geq 0$, and serves as a control that the engine never manufactures a certificate for a false statement. For the prefix series, which *is* coefficientwise nonnegative as far as we compute, it is informative: it extends [H, §10.3] from order 1 to order 7. The bound sharpens with probe depth: depth 1 gives ≥ 4 , depth 2 gives ≥ 7 , depth 3 rules out everything up to 7.

6 The unpaired tail

6.1 One series

Set $H(a) = (1 - a)/(1 - 2a)$ and $J_d = F_d/H$. After H is removed, every occurrence of a in a numerator or denominator monomial is accompanied by b : the chamber monomials containing a are $z_1/z_l = ab \cdots$, and only $z_1/z_2 = a$ is bare, which is exactly the factor $1 - 2a$ that H removes. Hence $\deg_a [b^j c^k \cdots] J_d \leq j$, and convolving with $H(a) = 1 + \sum_{n \geq 1} 2^{n-1} a^n$ gives, for $i > j$,

$$A_{i,j,k,\dots} = 2^{i-1} [b^j c^k \cdots] J_d(1/2, b, c, \dots).$$

The entire unpaired tail, infinitely many coefficients, is therefore one positivity question about a single series in $d - 2$ variables. This is [H, Lemma 2 and Prop. 3] for $d = 5$, stated so that it applies verbatim for any d .

Our $J_5(1/2, b, c, e)$ agrees with [H, eq. (9)] coefficient for coefficient to degree 12, which is what validates the construction. $J_6(1/2, b, c, d, e)$ is coefficientwise nonnegative as far as we compute ($\deg \leq 10$).

6.2 Multiplicative certificates

The two successful proofs in the literature, [N, Thm. 1] and [H, Prop. 3], are *multiplicative*: products of ratios $(1 - u)/(1 - v)$, each nonnegative by Lemma 1 because $v - u \geq 0$. That shape is not a finite sum $\sum_S P_S \prod_{r \in S} (1 - f_r)$ with $P_S \geq 0$, so the additive certificates of §4 structurally cannot find them.

The additive search is obstructed on $J_5(1/2, \cdot)$, whose nonnegativity is Proposition 3, so the additive class is not the one to use here.

Presenting the numerator in factored form, the Vandermonde part being already a factorisation, one can search for a matching of numerator factors to denominators satisfying the Lemma 1 condition, cancel whatever divides exactly, and ask whether what remains is nonnegative.

Proposition 10. *For $d = 5$ this closes: all nine numerator factors pair off, three denominators cancel exactly, and the leftover numerator is exactly 1. Proposition 3 is reconstructed mechanically.*

For $d = 6$ the same search does not close, and §4 says why that is not worth pursuing: the unpaired tail is a statement about individual A_β , and at $d = 7$ it is false. The mechanisation is therefore a way of reproducing and checking the small- d arguments of [H], not a route to a general one.

7 Assessment

What the computations suggest about proof strategy:

1. The τ -pairing reduction is *not* the right frame in general. Both its inequalities survive at $d = 6$ and both fail at $d = 7$ (§4), while Rimányi’s conjecture survives, so the machinery is specific to small d even though the statement is not. The prefix-positivity stepping stone fails one order earlier still.
2. Additive, Positivstellensatz-shaped certificates are obstructed on *every* target tried, including provable ones. On present evidence they are not the tool for this problem, and we would not recommend further effort extending [H, §10.3] in that direction.
3. Multiplicative certificates are the ones that succeed where anything does: they reproduce both published proofs from scratch. But what they certify — the individual A_β — is exactly what stops being controllable at $d = 7$, so they mechanise the small- d arguments rather than generalise them.
4. Since the τ -orbit is not the right unit of cancellation, the open question is what is. Table 5 is suggestive: the negative mass is small, bounded and spread over few orderings, while the positive mass is large and concentrated in one. A bound on the negative part of $C(M)$ in terms of its largest term would settle the conjecture without needing any pairing at all, and nothing we have computed argues against one existing.

8 Open questions and sharp edges

Points where these computations run past what we can justify from the literature. Each states what was checked, what it settles, and what it does not. Intended for discussion rather than as claims.

8.1 What Q_d is an invariant of

Take the corank-two 2-jet space $\text{Hom}(\mathbb{C}^2, \text{Sym}^2 \mathbb{C}^2)$, of dimension 6, and the class $I_{2,2}$ with local algebra $\mathbb{C}[[x, y]]/(xy, x^2 + y^2) \cong \mathbb{C}[[x, y]]/(x^2, y^2)$. Four germs of this one class give three different $B_2 \times B_2$ -orbit closures (Table 6); the two standard presentations of the *same* algebra already disagree.

The natural repair — demand the representative be generic against the flags — fails, for a structural reason. On this space $GL_2 \times GL_2$ *does* have a dense orbit, so $I_{2,2}$ is the generic corank-two 2-jet and has codimension 0; but $B_2 \times B_2$ does not, its generic orbit being 5-dimensional in a 6-dimensional space. The action has modality at least one, the generic jet lies

Table 6: Borel orbit closures of four germs of the single class $I_{2,2}$.

germ	orbit closure	codim
$(xy, x^2 + y^2)$	q_{02}^u	1
$(x^2 + xy, y^2)$	q_{02}^u	1
(x^2, y^2)	q_{02}^u, q_{11}^u	2
(y^2, x^2)	three quadrics, determinantal	2

on a one-parameter family of orbits, and their closures are hypersurfaces whose equations move with the representative. There is no generic Borel orbit to single out.

This is not a defect in [BS]. Their $\mathcal{Q}_d$ is not “the class of the A_d locus”; it is the equivariant class of *one specific* Borel orbit closure, that of a canonically defined reference jet, and the residue formula is what converts it into a Thom polynomial. The choice of ϵ_{ref} is part of the theorem, not a normalisation one is free to vary.

Question. Is there a corank-two analogue of ϵ_{ref} , together with a residue formula proved against it? Does the fibration of [BS, Lemma 7.1] have a corank-two counterpart? Without both halves a corank-two multidegree is a number with no theorem attached, which is why `multidegree/corank2.py` stops there and emits no artifact.

8.2 The acting group beyond 2-jets

On 2-jets only the *linear* parts of a source and target diffeomorphism act: for f with no constant or linear term, $f(\varphi_1 + \varphi_2 + \cdots) = f(\varphi_1) + O(3)$ and $\psi(f) = \psi_1(f) + O(4)$. From order 3 the higher parts contribute, so the acting group is the jet group. Every $I_{a,b}$ with $b \geq 3$ lives in the b -jet space — $I_{2,3}$ and $I_{3,3}$ in dimension 14, $I_{2,4}$ in dimension 24 — so the whole family past $I_{2,2}$ needs this.

Question. What is the right Borel of the jet group, and does the defect-zero saturation trick of §2.2 have an analogue that isolates the orbit component?

8.3 Which positivity, in which basis

Rimányi’s conjecture is a statement about the relative *Chern monomial* basis, and it is a corank-one statement. From Rimányi’s registry, at relative dimension 0, $\text{Tp}(I_{2,2}) = c_2^2 - c_1 c_3 = s_{2,2}$, the Giambelli–Thom–Porteous class of Σ^2 . We expanded the published $I_{2,2}$, $I_{2,3}$, $I_{2,4}$, $I_{3,3}$ in the Schur basis: every one is Schur-positive, and every one has mixed signs in Chern monomials. So the I family tests Schur positivity — Pragacz–Weber — and refutes any naive extension of the Chern-monomial statement.

Question. Is Chern-monomial positivity special to Morin singularities, or the shadow of a statement about a basis interpolating between Chern monomials and Schur polynomials? The A_d Thom polynomials are positive in both; the $I_{a,b}$ only in one.

8.4 Why $\mathcal{Q}_8$ needs different mathematics

Between $d = 7$ and $d = 8$ the ambient dimension goes $34 \rightarrow 50$, $\deg \mathcal{Q}_d$ goes $13 \rightarrow 22$ and the defect-zero coordinates $12 \rightarrow 16$. The obstruction is the very first Gröbner basis, and it is not a constant-factor problem.

The ideal is homogeneous, so a degree-bounded basis is exact up to its bound and walking the bound prices each degree: for A_8 the cost multiplies by between four and eight per degree, reaching degree 9 at 4 GB. The bound that would be needed is the other half of the estimate. The saturated basis reaches degree 3 at $d = 5$, 6 at $d = 6$ and 14 at $d = 7$, so $d = 8$ should want somewhere near 25 — some 10^{11} times the cost of degree 9, with memory giving out long before

time does. Ten variable orders and two algorithms all fail to finish that first basis; the base field is irrelevant; and seeding the computation with the already-saturated $\mathcal{Q}_7$ ideal, legitimate because truncating jets maps $\mathcal{O}_8$ into $\mathcal{O}_7$, does not help.

So the question is not how to speed this up but whether $\mathcal{Q}_8$ is needed. For Rimányi’s conjecture at $d = 8$ it is not: the registry publishes Tp_{A_8} and Tp_{A_9} at relative dimension 0, computed by restriction equations rather than by a residue formula, and all 22 and 30 coefficients are positive — the conjecture is supported two orders beyond where we compute. What $\mathcal{Q}_8$ would buy is what Chern coefficients cannot give: the individual A_β , and so the ability to ask at $d = 8$ what §4 answers at $d = 7$. That argues for a different route to $\mathcal{Q}_d$ — localisation on a resolution, or the fibration of [N, Lemma 7.1] — rather than a faster Gröbner basis.

Reproducibility

Everything above is reproducible from a clean checkout of Chern++. Statements are separated throughout into finite exact verifications and proofs; the certificate machinery and the closed-form family arguments are the only components producing the latter. The annotated walkthrough is `src/examples.ipynb`, executed with its outputs.